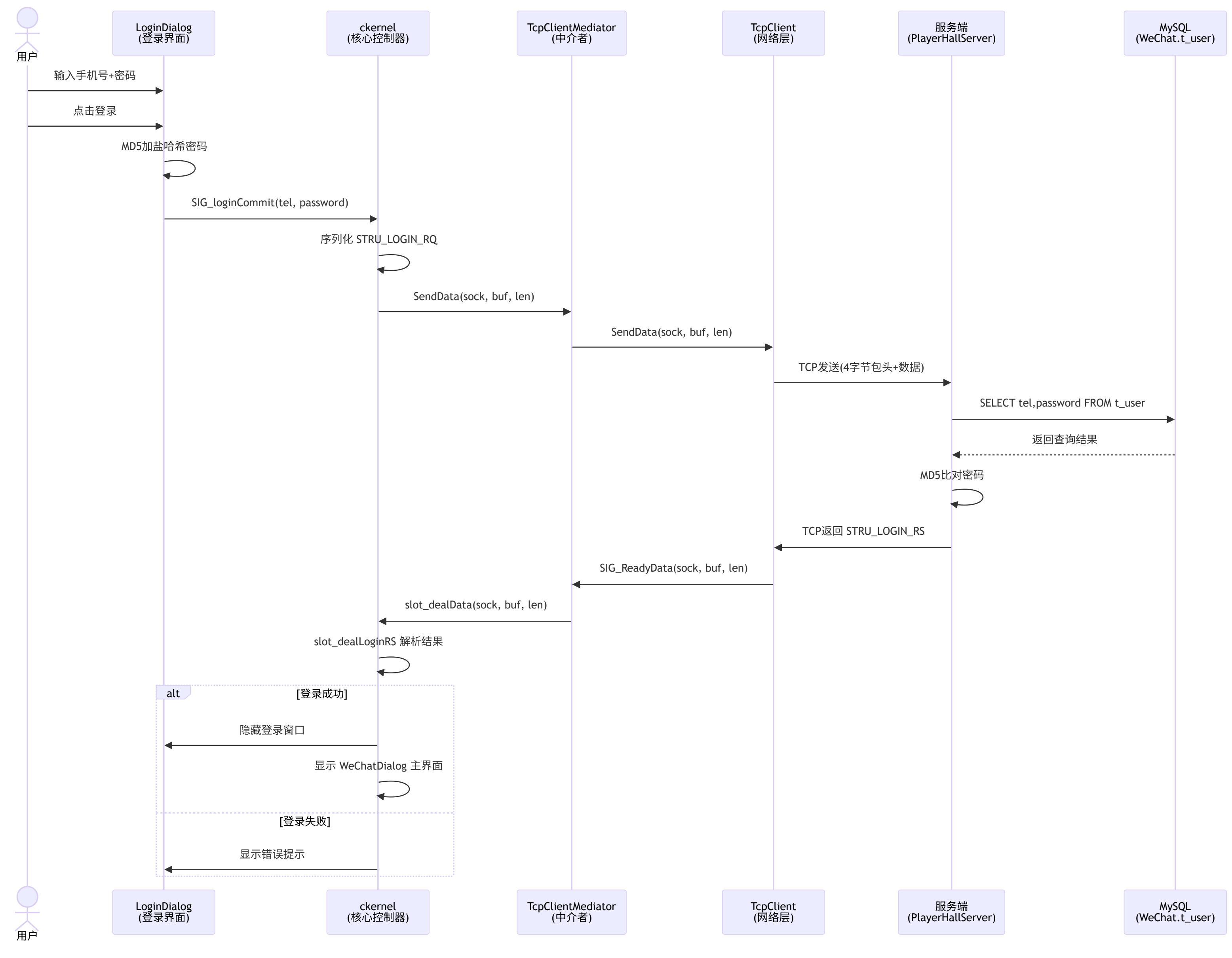
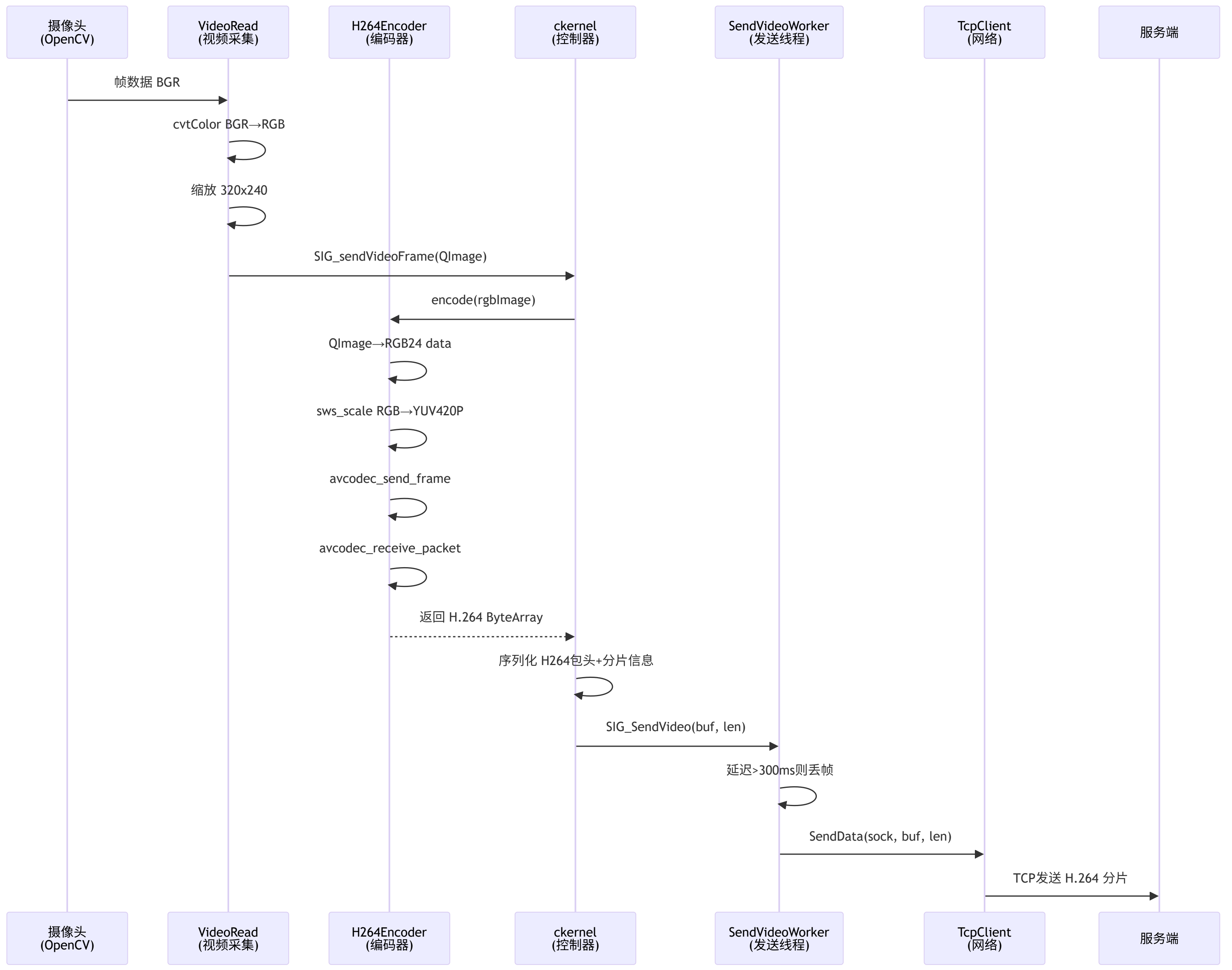
WeChat 音视频会议系统 - 系统时序图

展示系统核心交互的时序关系，覆盖登录、视频通话、音视频同步等关键流程

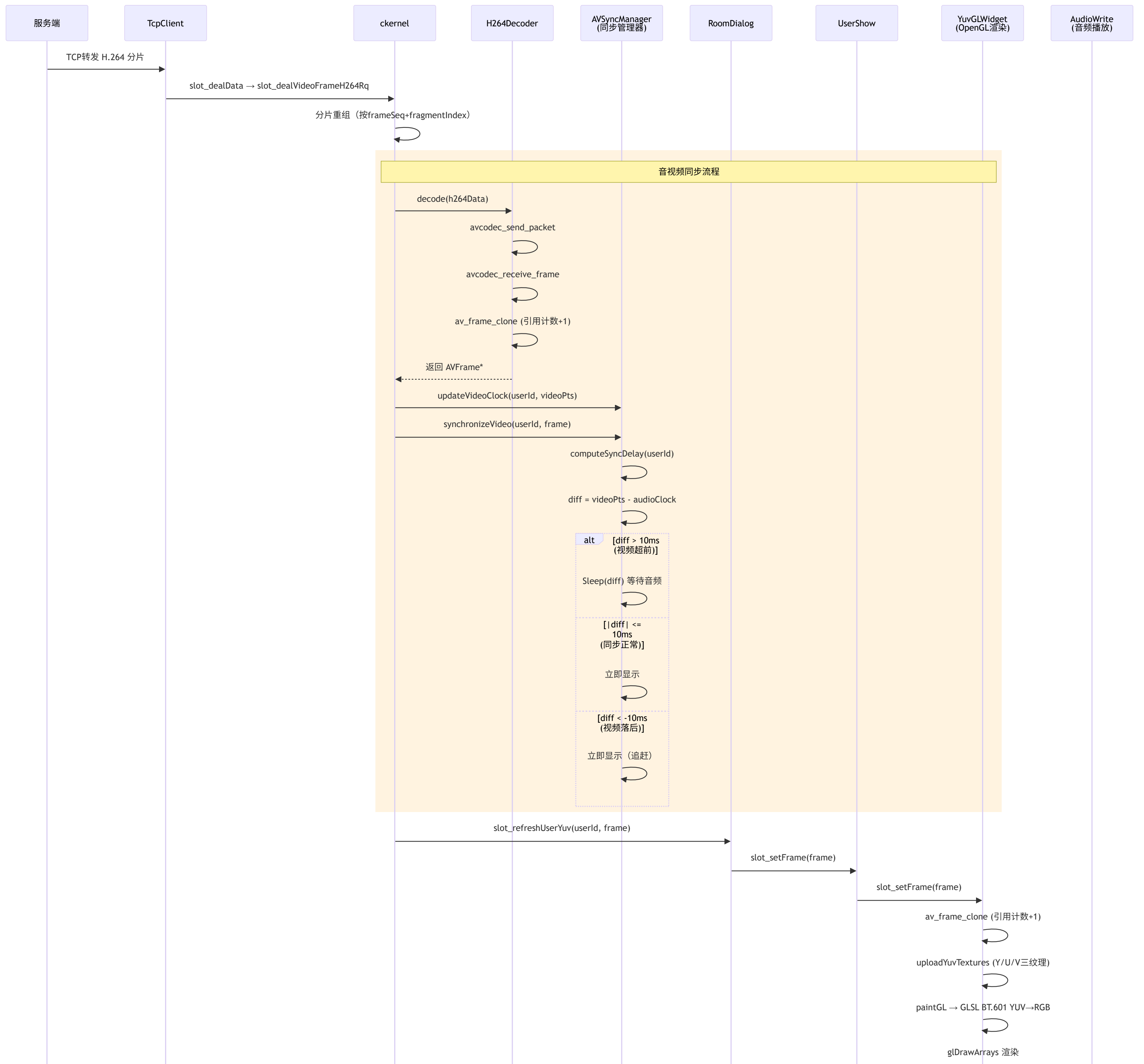
一、用户登录时序图



二、H.264 视频通话时序图



三、接收端 H.264 解码+同步时序图


服务端
TcpClient
ckernel
H264Decoder
AVSyncManager (同步管理器)
RoomDialog
UserShow
YuvGLWidget (OpenGL渲染)
AudioWrite (音频播放)
TCP转发 H.264 分片
slot_dealData → slot_dealVideoFrameH264Rq
分片重组（按frameSeq+fragmentIndex）
音视频同步流程
decode(h264Data)
avcodec_send_packet
avcodec_receive_frame
av_frame_clone (引用计数+1)
返回 AVFrame*
updateVideoClock(userId, videoPts)
synchronizeVideo(userId, frame)
computeSyncDelay(userId)
diff = videoPts - audioClock
alt
[diff > 10ms (视频超前)]
Sleep(diff) 等待音频
[|diff| <= 10ms (同步正常)]
立即显示
[diff < -10ms (视频落后)]
立即显示（追赶）
slot_refreshUserYuv(userId, frame)
slot_setFrame(frame)
slot_setFrame(frame)
av_frame_clone (引用计数+1)
uploadYuvTextures (Y/U/V三纹理)
paintGL → GLSL BT.601 YUV→RGB
glDrawArrays 渲染
音频独立时钟更新
Opus解码 → SDL播放
updateAudioClock(userId, audioPts)

四、新成员加入房间时序图